

Danny Rollo

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Available now

RESEARCH INTERESTS

Deep learning for medical imaging and healthcare; machine learning for urban planning and public transit; computer vision and representation learning for football (soccer); recommender systems and cold-start generalization; imitation learning for robotic manipulation; agentic LLM systems.

EDUCATION

Northeastern University

M.S. in Computer Science

Boston, MA

Sept 2024 – April 2026

B.S. in Computer Science & Mathematics

Sept 2021 – May 2025

- *Graduate coursework:* Applied Deep Learning, Deep Learning, Machine Learning, Algorithms, Empirical Research Methods, Web Development, Computer Graphics
- *Undergraduate coursework:* AI4Impact, Matrix Methods in Machine Learning & Data Analysis, Macroeconomics, Systems Security, Object-Oriented Design, Statistics & Stochastic Processes

PUBLICATIONS

Peer-Reviewed Conference

1. **Rollo, D.** “Lightweight CNN Encoders for Nucleus Segmentation.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026. **Accepted — poster.**

A controlled encoder ablation showing a 34M-parameter VGG U-Net matches MedT — a transformer purpose-built for small medical datasets — at a $\sim 24\times$ smaller parameter budget, and outperforms pretrained Swin-T by 4.9 Dice points with $2.6\times$ less memory and $1.8\times$ lower inference latency.

In Preparation

2. **Rollo, D.**, et al. “Two-Tower Recommendation with Cold Start Ablation.” *Manuscript in preparation.*

A 12-experiment ablation on a dual-encoder retrieval model over 1.3M Yelp reviews, with a leave-one-out onboarding protocol that prevents test-set leakage on cold users.

RESEARCH EXPERIENCE

Lightweight CNN Encoders for Nucleus Segmentation

2025–2026

Sole author · Northeastern University

PyTorch, timm, PanNuke, MoNuSeg

- Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
- Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.
- Attributed the result to spatial-resolution preservation at skip connections and convolutional inductive bias outweighing Swin’s pretraining advantage in the low-data regime.

- Authored the full manuscript, produced all quantitative and qualitative figures, and wrote the discussion.

Two-Tower Recommendation with Cold Start Ablation

2025

Primary contributor & project lead · team of 4

PyTorch, BPR loss, Yelp Open Dataset

- Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
- Designed a leave-one-out onboarding evaluation that prevents information leakage when a cold user's only signal comes from held-out reviews.
- Built a 4-stage data pipeline (extract, preprocess, split, feature prep) with iterative interaction-density filtering and stratified cold-start holdout; reduced dataset construction from 47s to 8s via vectorization.
- Implemented on-device feature indexing eliminating per-batch CPU–GPU transfer, with per-epoch resampling of geographic negatives and Gumbel-max onboarding selections as data augmentation.

Flow Matching for Robotic Manipulation

2025

Team of 3 · Northeastern University

PyTorch, MuJoCo, MetaWorld

- Implemented continuous-time flow matching for imitation learning: a neural vector field transports Gaussian noise to expert action distributions via RK2 ODE integration, avoiding the iterative denoising cost of diffusion policies.
- Scaled from a proprioception-only baseline on single-object bin-picking to an RGBD- and language-conditioned policy with 8-step action chunking across 3-object scenes; characterized compounding-error limits on multi-object generalization.

RailMax: Amtrak Service Optimizer

2025

Open-source modeling study

InterpretML, GTFS, Census ACS, Folium

- Modeled the US Amtrak network as a directed weighted graph (edges = weekly trips between station pairs) and engineered node-level features including destination-population-weighted reachability and eigenvector centrality.
- Trained an Explainable Boosting Machine ($R^2 = 0.76$, RMSE 0.84 log scale) under 5-fold stratified group cross-validation with k-means geographic clustering to prevent spatial leakage.
- Integrated Census ACS, GTFS, NTAD, and IPEDS sources; deployed an interactive what-if frequency simulator identifying the top-20 city pairs most likely to benefit from added service.

NJ Transit Reliability Tracker: Independent OTP Measurement

2026

Sole author · live public deployment

Cloudflare Workers, GTFS-Realtime, SQL, TypeScript

- Built and deployed a live public reliability tracker for NJ Transit commuter rail on Cloudflare Workers, continuously ingesting GTFS-Realtime arrivals and recomputing on-time performance at stricter thresholds than the agency's published 6-minute definition (88% official vs. 80.6% measured at 5 minutes).
- Reported full delay distributions, time-of-day and day-of-week heatmaps, and transfer-connection reliability in place of a single headline rate, with a significance gate suppressing spurious week-over-week movements on low-frequency lines.
- Unified three sources of differing cadence — GTFS static, monthly published performance reports back to 2017, and the realtime feed — with per-figure provenance labelling separating official from independently measured numbers.

Hopewell Road Sketches: Parametric Street-Safety Modeling

2026

Sole author · open source

Python, geopandas, shapely, trimesh, Blender

- Built a parametric geometry pipeline (geopandas, shapely, trimesh) that generates to-scale 2D and 3D before/after street-design comparisons for real intersections from NJDOT road data, OpenStreetMap geometry, and field measurements.
- Computed safety outcomes from first principles — crossing distance, pedestrian time-in-roadway at MUTCD 3.5 ft/s, AASHTO turn speed at tightened corner radii, and parking loss from paint geometry — with every input traced to an authoritative source rather than a geometric default.
- Enforced cross-representation consistency with regression and property-based tests, surfacing geometry defects between the 2D and 3D pipelines that visual inspection had missed.

Hopewell Area Crash & Crosswalk Map

2026

Sole author · open source, in public use

Leaflet, Python, OpenStreetMap, GitHub Pages

- Built an interactive crash and pedestrian-infrastructure map for Hopewell Borough, NJ, combining incident records with a personally conducted field survey of 84 crosswalks across 44 locations.
- Designed a repaving priority score integrating crash-harm proximity, survey condition grades, and municipal paving cycles, and derived crossing geometry from OpenStreetMap with a road-centerline fallback for unmapped crossings.

Booker

2025

Sole author · live demo

Python, Claude API, MongoDB Atlas Vector Search, MCP

- Built a multi-agent system for artist–venue booking matching: a coordinator agent with bounded iterations routes to specialist agents (artist, venue, booking advisor) via structured, typed tool calls against the Claude API, preventing runaway agent loops by construction.
- Implemented semantic artist/venue discovery over MongoDB Atlas Vector Search using `all-mpnet-base-v2` embeddings (768-dim), with a 3-tier memory system (conversation, working, preferences) for context-aware multi-turn interactions.
- Enforced content safety and domain boundaries with NeMo Guardrails, exposed the system via an MCP server for standardized Claude Desktop tool interop, and instrumented it with OpenTelemetry tracing.

PROFESSIONAL EXPERIENCE

Pitchcut — Founder & CTO

June 2026 – Present

Remote

- Architected a multi-tenant SaaS platform for grassroots football clubs (Postgres/Supabase, row-level tenant isolation) combining CV-based match analytics with an agentic content-generation pipeline.
- Built a computer-vision pipeline (tracking, event/goal detection, player re-identification) validated against SoccerTrack v2 ground truth, plus a consent-gated data flywheel for training-set collection.
- Designed an agent-driven back office (Claude Code skills over a typed domain layer) automating registration, roster, and content drafting, with human-in-the-loop approval for all public-facing actions.
- Optimized the CV pipeline’s CUDA inference path (single-GPU-upload tile cropping for 17-tile SAHI-style detection, fp16 autocast) for a $2.4\times$ throughput gain (245ms $\rightarrow$ 102ms/frame) over the CPU-tile fp32 baseline, auto-detected across CUDA/MPS/CPU and parity-verified before shipping.
- Built submitit-managed SLURM training infrastructure under Explorer’s single-GPU-per-job cap: throttled job arrays with checkpoint/resume, node-throughput profiling that auto-excludes slow nodes (26.8 img/s V100 vs. 3.75 T4, a $7.1\times$ gap), OAuth-storm-tuned retry/backoff (526 errors in ~ 16 s), and MLflow logging that survives SIGKILL-on-timeout.

Teradata — AI/ML Product Management Analytics Intern
San Diego, CA (Remote)

June 2025 – May 2026

- Authored the business case and led the design of a context-aware AI Notebook Assistant for Vantage-Cloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
- Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.

PlateMate — Co-Founder & Chief Technology Officer
Boston, MA

May 2025 – Present

- Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus semantic vector search over restaurant reviews.
- Built a RAG retrieval pipeline with 768-dimensional `all-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.

LineVision Inc. — Software Engineer Co-op
Boston, MA

Jan 2024 – June 2024

- Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
- Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.

Brekinridge Capital Advisors — Infrastructure Intern
Boston, MA

July 2023 – Sept 2023

- Cleaned datasets, built Python web scrapers, and trained a clustering model on the resulting data.
- Performed Windows and Linux system administration and security work using PowerShell, Darktrace, and Bit9; built front-ends in React.js and Razor Pages.

Generate Product Development
Boston, MA

Jan 2023 – May 2025

Project Lead

Dec 2024 – May 2025

- Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.

Technical Lead

Sept 2023 – Dec 2023

- Led a team of developers building an iOS/Android app on a React Native frontend with a Go/Postgres backend.

Software Engineer

Jan 2023 – May 2024

- Backend developer on a team of 8 building ReMo, a literacy tool for educators; designed the database and internal model and implemented the networking API in Go.

TEACHING EXPERIENCE

Khoury College of Computer Sciences — Lead Teaching Assistant
Fundamentals of Computer Science I & II

2022 – 2025
Boston, MA

- Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.

- Taught recursion, linked lists, and functional programming (Fundamentals I), then Java object-oriented design, polymorphism, data structures, algorithm design, and time complexity (Fundamentals II).

HONORS & AWARDS

Rookie of the Year, HackBeanpot

February 2023

Built a food-desert tracking web application on a six-person team.

Students of the Year Runner-Up, Leukemia & Lymphoma Society

March 2021

Raised \$58,000 for blood cancer research in seven weeks during the height of the COVID-19 pandemic.

Eagle Scout, Scouts BSA

October 2019

Led 30 volunteers to design, fundraise, and construct six portable benches for a local school, delivered on schedule despite pandemic restrictions and supply shortages.

TECHNICAL SKILLS

Languages & Core Libraries: Python, Go, TypeScript/JavaScript, SQL, pandas, NumPy

Deep Learning: PyTorch, timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention, flow matching, diffusion policies, imitation learning

Machine Learning: scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering, two-tower retrieval, ablation design, cross-validation methodology

LLM & Agentic Systems: Claude and OpenAI APIs, MCP servers, multi-agent orchestration, prompt engineering, RAG, SentenceTransformers, vector search, NeMo Guardrails, OpenTelemetry

Infrastructure: Docker, AWS (S3), CUDA, SLURM job arrays, submitit, MLflow, FastAPI, MongoDB Atlas, Git

Statistical & Research Computing: Stata (coursework), R (basic), scipy, LaTeX, Unix/Linux, SLURM / HPC clusters

Data & Visualization: Folium, Streamlit, GTFS, Census ACS

LANGUAGES & ACTIVITIES

Languages: Latin (limited working); English (native).

Activities: Generate Product Development (Northeastern student-run product studio); captain, Northeastern FC (BSSL) and intramural soccer; guitarist and songwriter, Northeastern Jazz Combo and Music Farm.

References available upon request.